

Complex Analysis PhD Exam, May 2000

Notes. Please write up solutions to the eight problems 1–8, below. Please write LARGE, as the grader's eyes are older and weaker than your eyes.

A square $S \subset \mathbb{C}$ is a set of the form $[a, a + L] \times [b, b + L]$, where L is positive. Use “ $s := \text{Foo}$ ” to mean that Foo is the definition of the new symbol s .

1. Find complex numbers a, b, c, d , with $ad - bc \neq 0$, so that the Möbius transformation

$$\mu(z) := \frac{az + b}{cz + d}$$

carries the imaginary axis to the circle whose radius is 2 and whose center is $3 = 3 + 0i$.

2. With B the open unit ball $|z| < 1$, consider a non-constant analytic function $h : B \rightarrow \mathbb{C}$.

- i: Suppose that $\Re(h(z)) \geq 0$ for each $z \in B$. Prove that the inequality can then be strengthened to “ $>$ ”.
- ii: With $\Re(h(z)) > 0$ on B , suppose further that $h(0) = 1$. Prove, for each $z \in B$, that

$$\frac{1 - |z|}{1 + |z|} \leq |h(z)| \leq \frac{1 + |z|}{1 - |z|}.$$

3. Suppose that $P()$ is a monic polynomial with degree $N \geq 1$. Let $\alpha_1, \dots, \alpha_N$ be an enumeration (with multiplicity) of the zeros of $P()$.

- ① Suppose that $\forall k : \Re(\alpha_k) > 0$. Prove that all the zeros of the derivative, P' , also lie in the positive half-plane, as follows: Establish that

$$\frac{P'(z)}{P(z)} = \frac{1}{z - \alpha_1} + \frac{1}{z - \alpha_2} + \dots + \frac{1}{z - \alpha_N},$$

then use it to complete the proof.

- ② Prove Lucas's theorem: If all the zeros of a non-constant polynomial P lie in a convex polygon $Q \subset \mathbb{C}$, then all the zeros of P' lie in Q .
- ③ Show that ② can fail if $P()$ is allowed to be a rational function: Namely, by letting

$$P(z) := \frac{z}{z^2 + 1},$$

find a half-plane H which owns a zero of P' but has no zero of P .

4. Use the Residue Calculus to compute

$$I := \int_0^{+\infty} \frac{1}{[x^4 + 4] \cdot [x^2 + 9]^9} dx.$$

To save arithmetic, you may define some explicit points $P_1, \dots, P_L \in \mathbb{C}$ (what should L be?) and explicit functions $h_1, \dots, h_L$, and then may express your answer in the form

$$I = [h_1(P_1) + \dots + h_L(P_L)] \cdot \text{Constant}.$$

(Do not bother to perform the function-evaluation.)

5. This problem has two parts.

- a: State (but do not prove) Morera's Theorem. (You may use this without proof in (b), if you wish.)
- b: Prove this version of the Schwarz Reflection Principle: Suppose f is continuous in the closed upper half-plane $H = \mathbb{R} \times [0, \infty)$ and is analytic on the interior of H . Further suppose that f is real-valued on the real-axis. By defining $\Phi = f$ on H , and

$$\Phi(z) := \overline{f(\bar{z})}, \quad \text{for all } z \in \mathbb{C} \setminus H,$$

extend f to all of $\mathbb{C}$. Then this Φ is analytic.

6. This problem has three parts.

- ① Please state Picard's Theorem.
- ② Let f be meromorphic in the whole complex plane. Suppose that the range of f omits three distinct values (one of them can be ∞). Prove that f is constant.
- ③ Suppose that f and g are entire functions such that, on $\mathbb{C}$,

$$f^3 + g^3 = 1.$$

Prove that f and g are each constant functions. [Note: Symbol f^3 means $f \cdot f \cdot f$.]

- 7. Fix a real $b > 0$. Write down an entire function, f , that vanishes precisely on the sequence $(z_n)_{n=1}^{\infty}$, where $z_n := n^b$.
- 8. Show that all roots of polynomial $P(z) := z^5 + 15z + 1$ lie in the ball $|z| < 2$, but that only one root satisfies $|z| < \frac{3}{2}$.